Data, Decisions, and AI/ML Discoveries: Lab B

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This is an optional lab, for which no credit will be given.

1 Objectives

On completing this project, you will be able to:

1. Load data from a text file
2. Visualize data with matplotlib
3. Create a k nearest neighbors model with Python
4. Create a decision tree with Python

2 Case Study

3 The Project

An example of an iris in bloom.

An iris is a kind of flower. You are given 150 rows of data. Each row represents information about a particular iris flower someone went and looked at.

The data is stored in a file called `iris_data.csv`. The first row is the header.

Recorded in the first column is the species of the flower. It is encoded here as a number: 1, 2, or 3.

Recorded in the second column is the average area of the flower's sepals in cm^2.

Recorded in the third column is the average area of the flower's petals in cm^2.

Our goal is, given sepal area and petal area, to predict if a new flower's species is 1, 2, or 3.

We will do so in a visual way: by plotting the decision boundary of a classifier.[a]

3.1 Deliverables

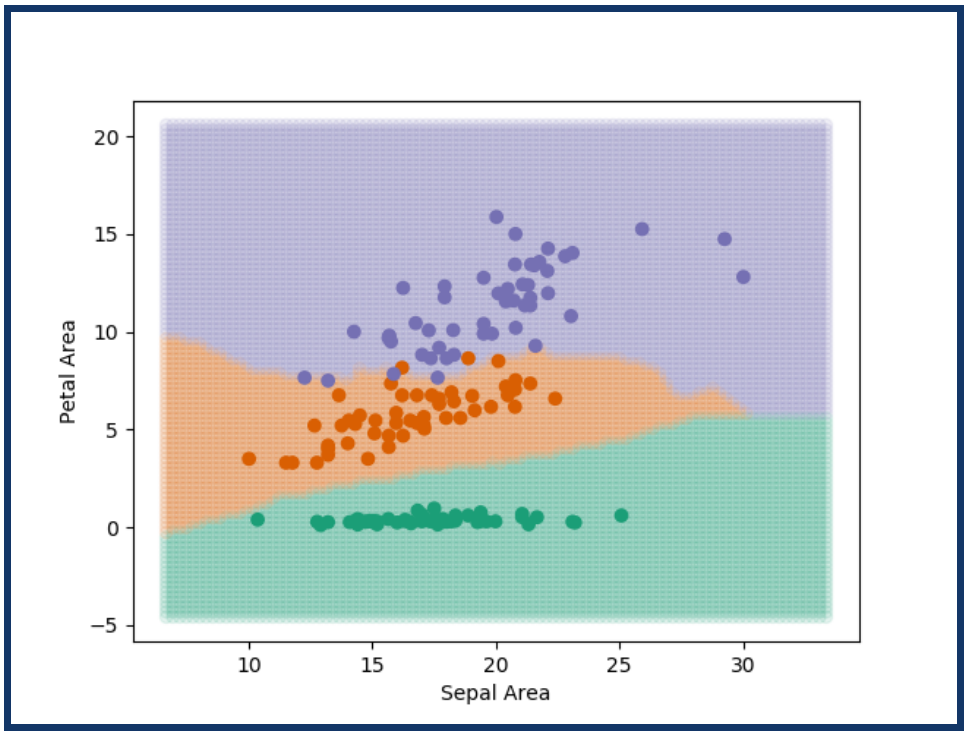
A Python program which:

1. Uses the sci-kit learn framework to classify data using at least two classifiers.
2. Plots the decision areas for each classifier.
3. Includes a scatter plot of the training data overlaying the decision area.
4. Displays these plots on the screen.

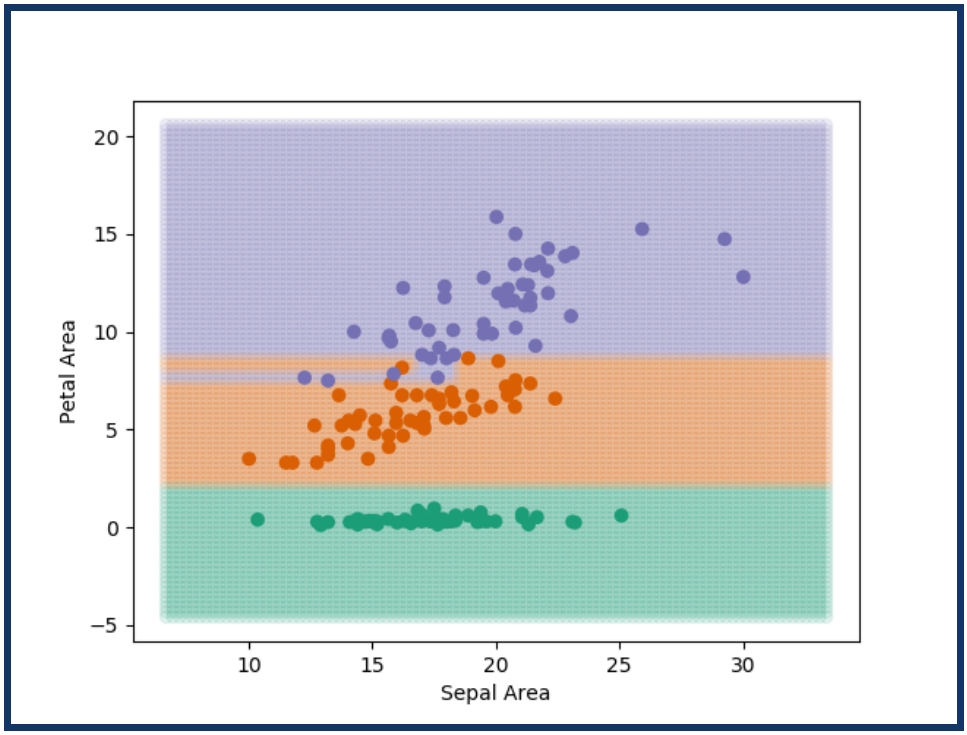
4 Ideal Output

Figures analogous to these:

[a] We will not be doing any validation on our classifier for this project. All data is to be used as training data.



Decision areas of a nearest neighbors classifier. Training data is overlayed.



Decision areas of a single decision tree. Training data is overlayed.